

The Qubit

An Educational Strategy Game Exploring Quantum Computing

KM5a

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Pitch Overview

- Main Concept
- Quantum Computing Integration
- Educational Purpose
- Conclusion and Future Vision
- Game Structure
- Development Milestones
- Testing and Feedback
- Q&A

Quantum Computing is...

Profitable

Change the future of technology.

BUT very abstract!

'Superposition', 'entanglement'...??

Traditional learning

Boring.

Taught interactively

A game.

The Qubit

A Tower Defense Game

Game Concept

Quantum Core

Protected by Qubits

OneQubit

It attacks

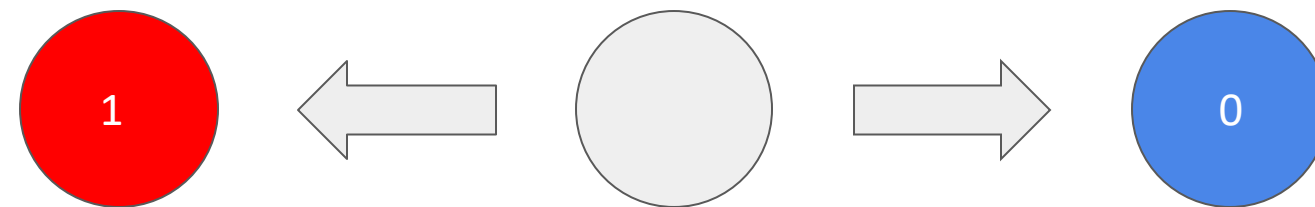
ZeroQubit

Generates Resources



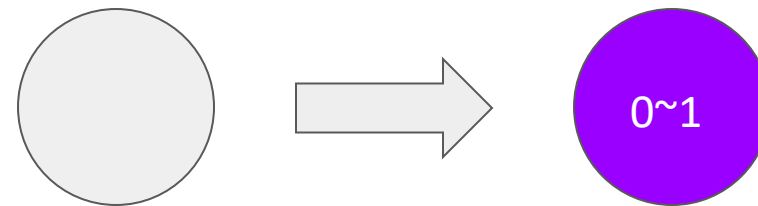
Quantum gates

- X gate change the type of qubit



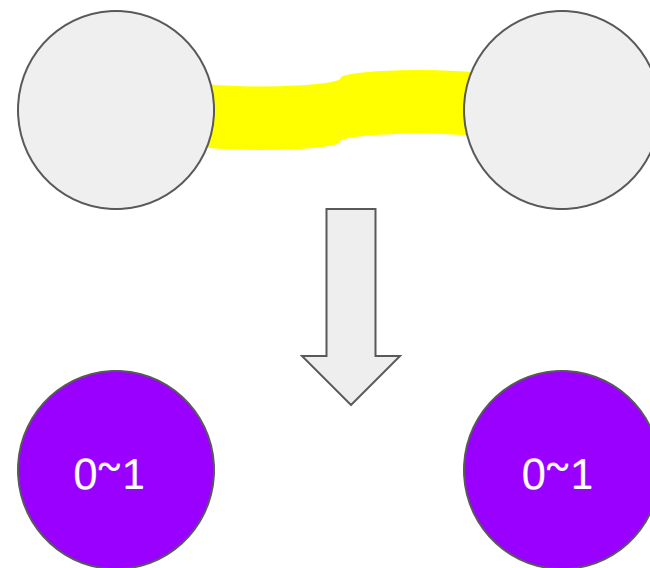
Quantum gates

- H gate change the qubit into superposition



Quantum gates

- Entanglement change multiple qubit at the same time



Other gates

Z Gate

Y Gate

S Gate

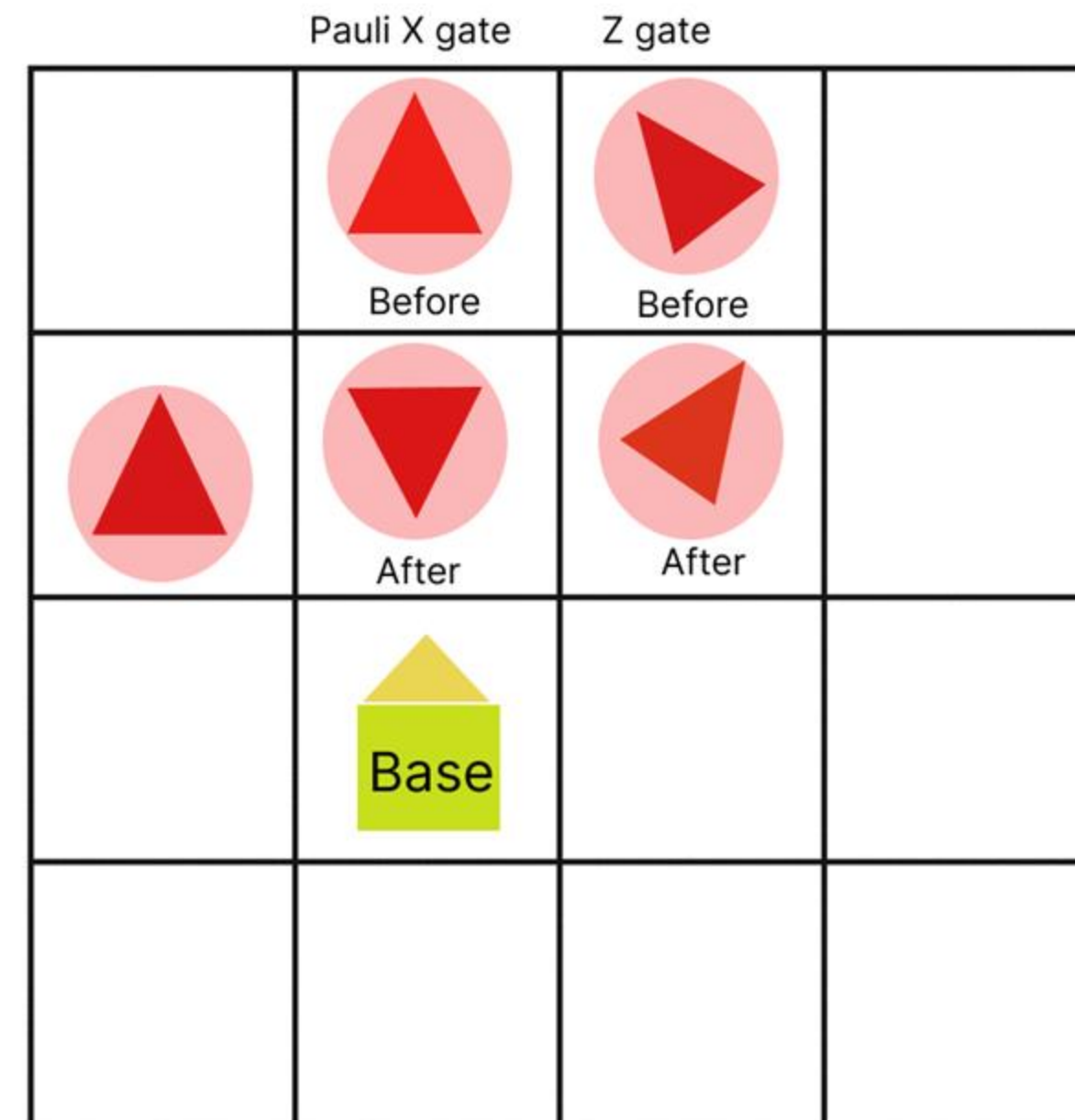
More other gates

Educational Applications of Quantum Computing

Quantum Gates:

- **Pauli X Gate:** *Rotates the Qubit 180 degree*
- **Z Gate:** *Phase flip*
- **H Gate:** *Creates Superposition*

Normal state

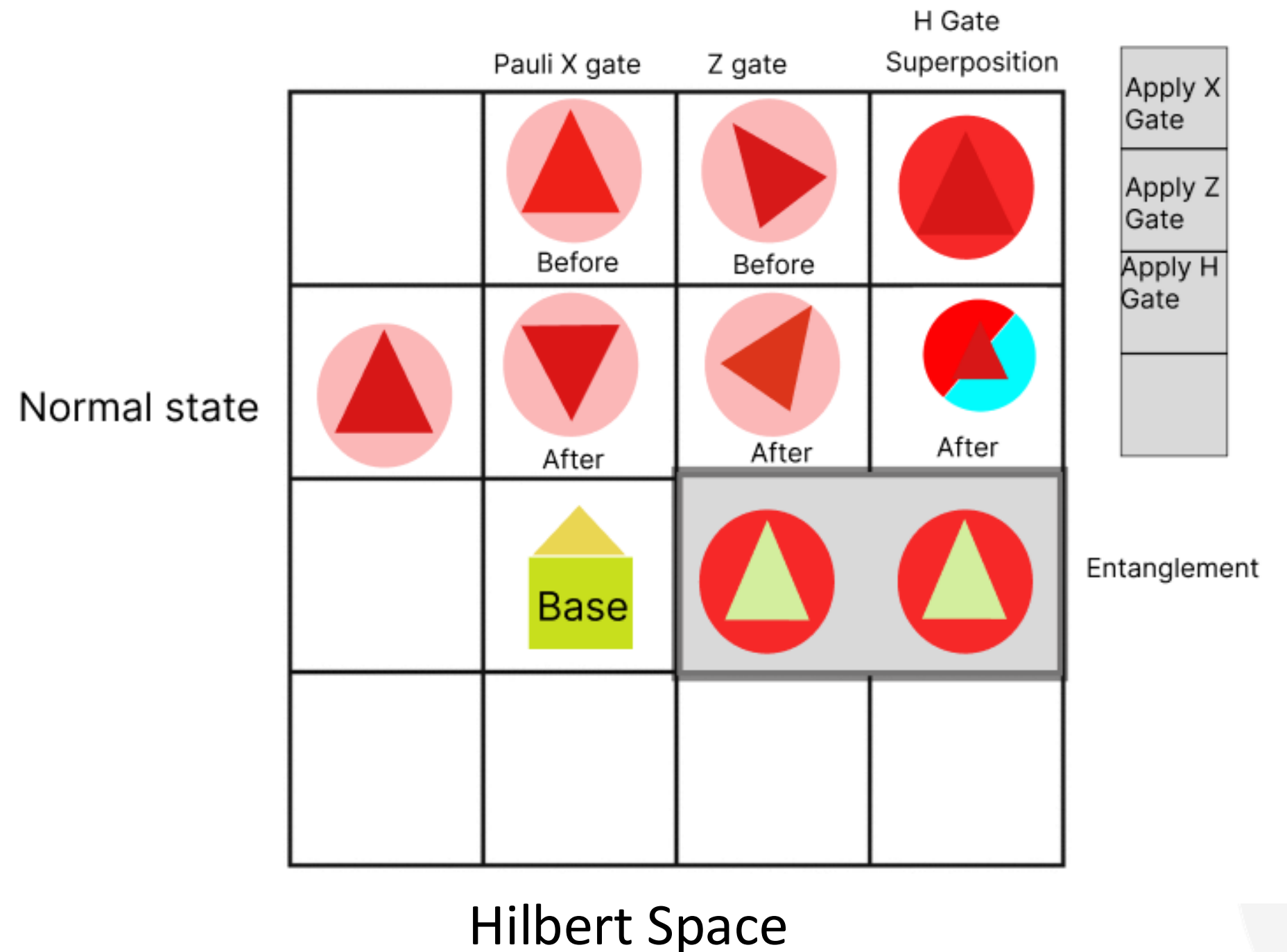


Hilbert Space

Educational Applications of Quantum Computing

Concepts:

- **Superposition** : *Simultaneous States*
- **Entanglement** : *Quantum Link*



Game Overview

- Core Game Systems
- Game Loop

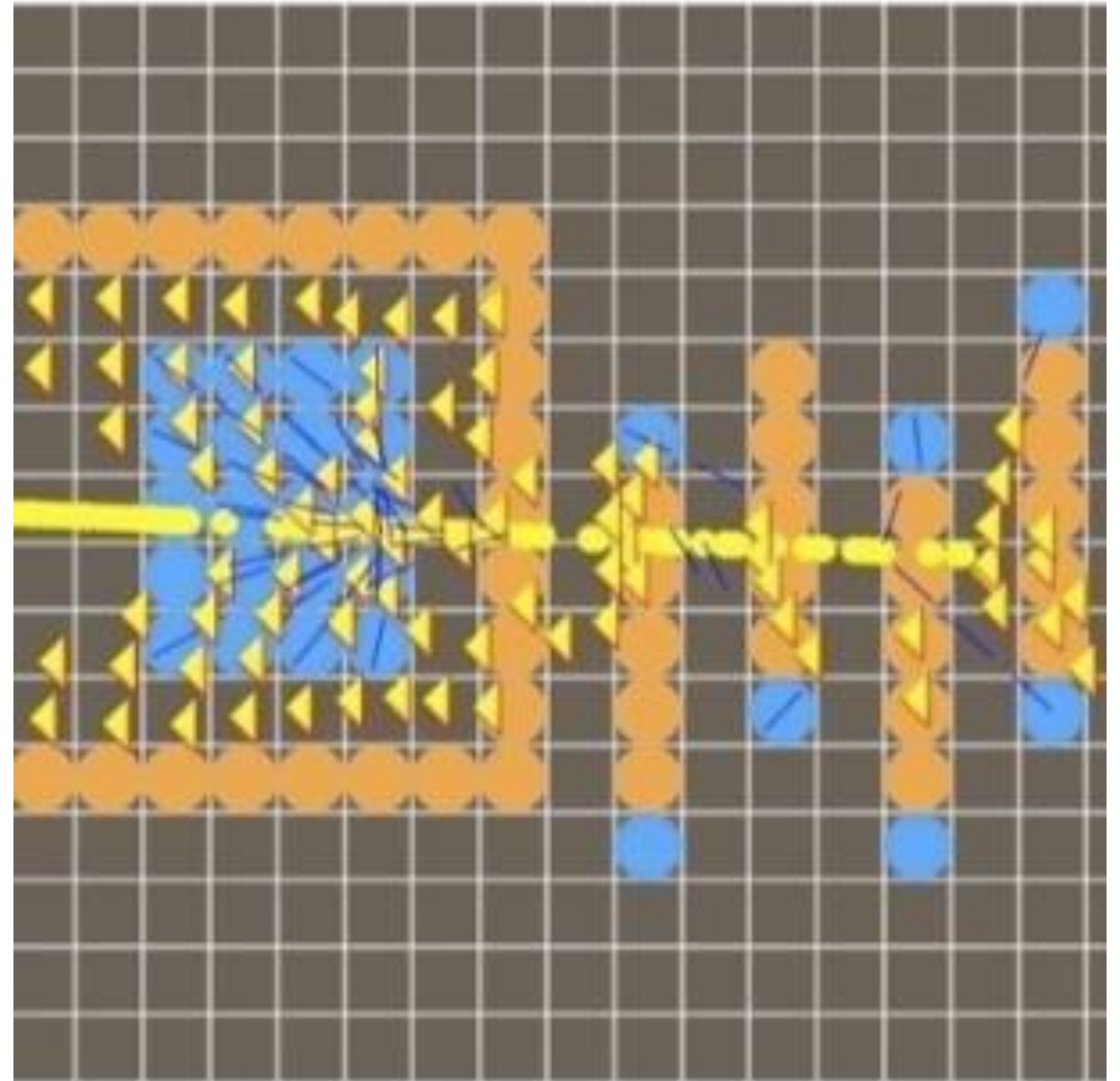
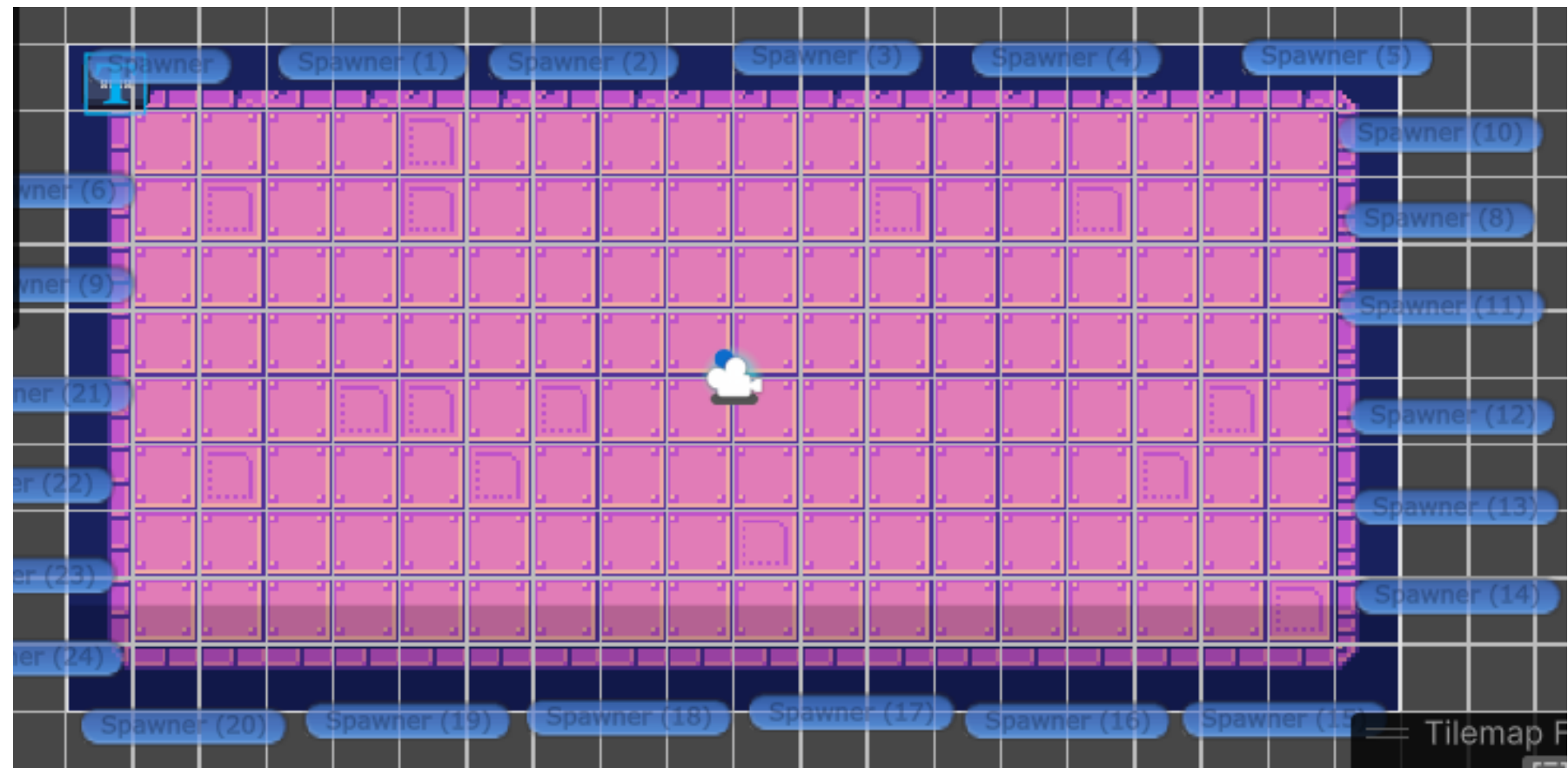
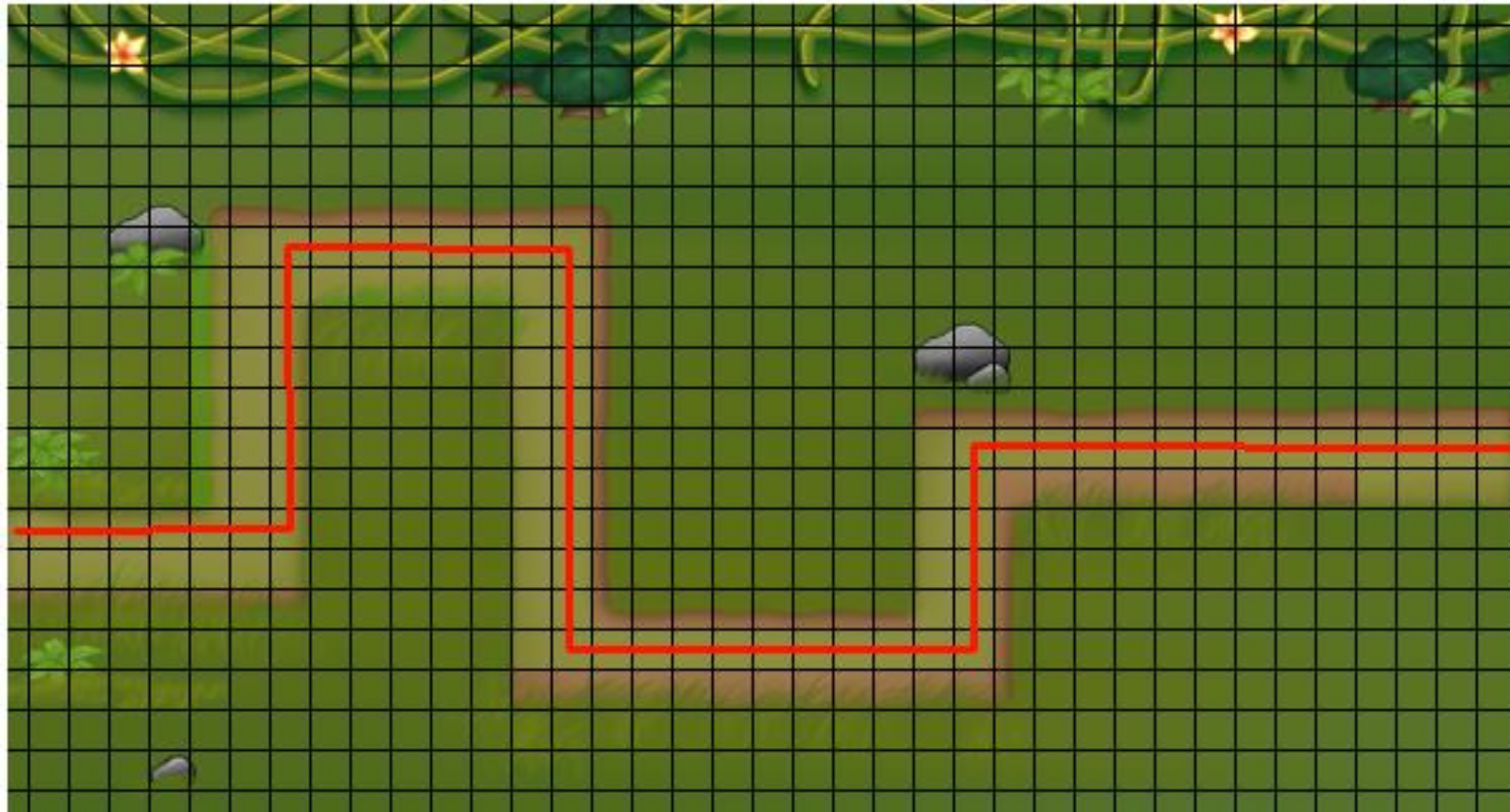
Core Game Systems

Grid System

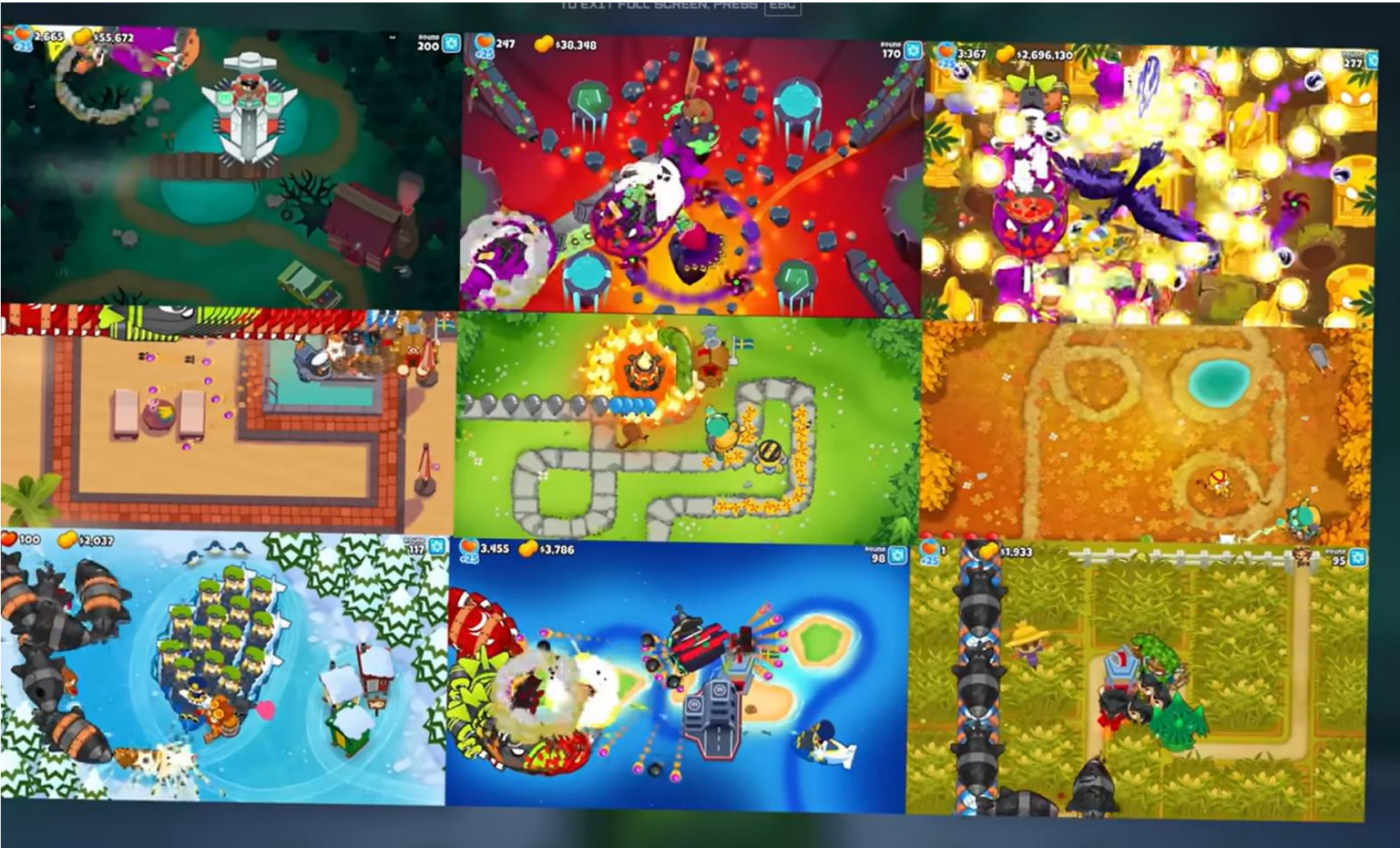
Wave Progression

Resource System

Rogue-like Progression







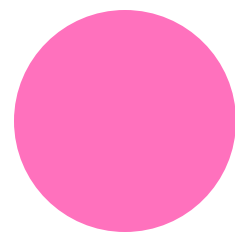


Gameplay Loop



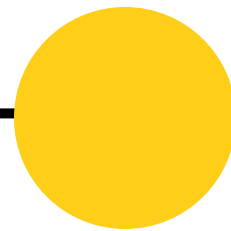
Milestone

Briefly discuss the key dates for the project.



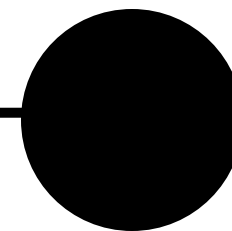
Week1-2 Planning & Research

Define gameplay and educational goals



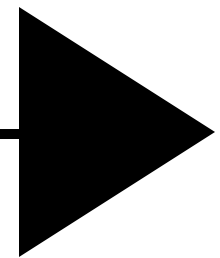
Week3-4 Core System Setup

Build basic playable framework



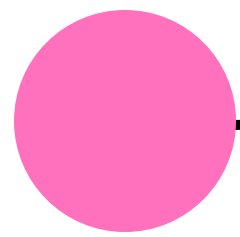
Week5-6 Gameplay Development

Polish core mechanics and integrate quantum logic



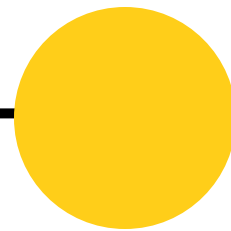
Milestone

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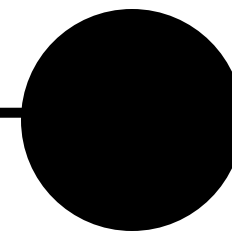
Week7-10 Educational Integration

Embed quantum concepts
into gameplay



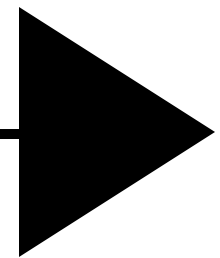
Week11 Testing & Balancing

Refine game experience



Week12 Final Presentation

Prepare demo & deliverables



Testing Strategy

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Haoyu Zhang



Parts of Testing Strategies

1. Unit Testing

2. User Testing

3. Integrate Testing

Unit Testing

What is it?

Verify Program components



Features

- 1. Testing a single module
- 2. Automatic operation

Expected Outcome

1. Improve code quality

Health System	Damage calculation	Health values update correctly
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User Testing

Test subjects

Student and teacher

We will:

11. Get user feedback

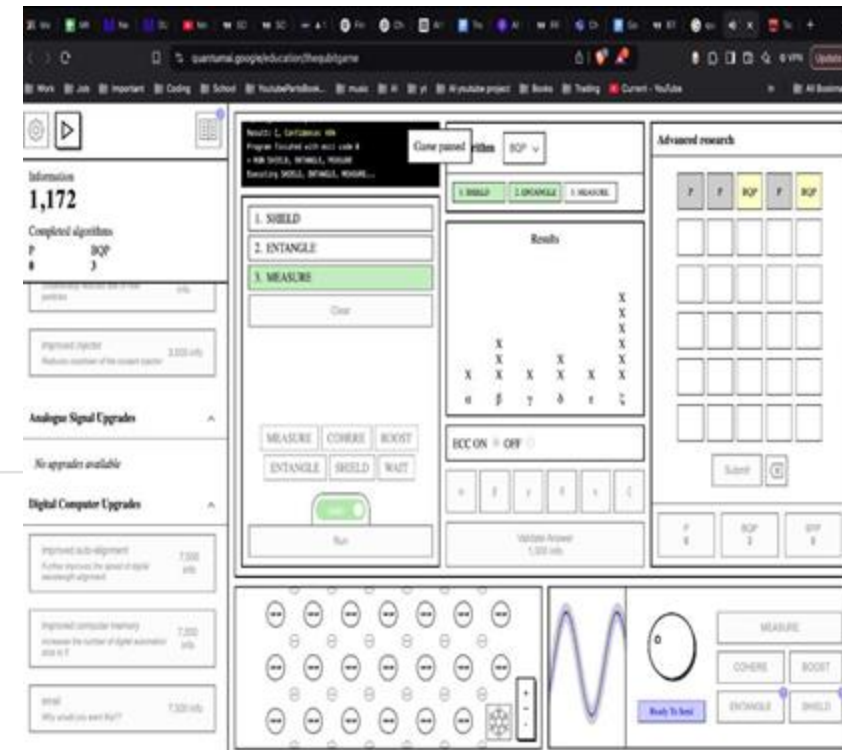


Integration Testing



1.What is it?

Interact properly



Function

1. Check data flow
2. Reduce bug

Why This Matters

1. Lowers the barrier to entry for quantum computing
2. Makes abstract concepts tangible and interactive
3. Attracts new students to quantum computing
4. Bridges the gap between gaming and education
5. Creates a foundation for future quantum literacy

Conclusion

- 1.itch.io platform
2. evaluate popularity

Thanks for your attention